MTH636 – Game Theory

Lecture 1: Strategic Thinking & the Beauty Contest Game

Dr. Soumyarup Sadhukhan | Dept. of Statistics & Data Science, IIT Kanpur

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1 Motivation: What Is Game Theory?

Game theory is the mathematical study of **strategic interaction** – situations where the outcome for each participant depends not only on their own actions, but also on the actions of others.

Intuition

Optimization vs. Strategy. In a standard optimization problem (e.g., calculus, operations research), you maximize/minimize a function that depends only on your own choices. In a *strategic* problem, your payoff depends on what *everyone else* does – so you must think about what others will do, which depends on what they think *you* will do, and so on. This feedback loop is the heart of game theory.

2 The Beauty Contest Game (Keynes' Guessing Game)

2.1 Setup

Definition: The p-Beauty Contest Game

- n players, each independently chooses a real number $x_i \in [0, 100]$.
- After all choices are made, compute the **average**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- The **target** is:

$$T(\mathbf{x}) = \frac{2}{3}\,\bar{x}$$

- The **winner** is the player whose choice x_i is closest to $T(\mathbf{x}) = \frac{2}{3}\bar{x}$.
- (In the generalized version: target $= p\bar{x}$ for parameter $p \in (0, \infty)$.)

Note

This is a *simultaneous* game – no player sees others' choices before deciding. It is *not* a pure optimization problem because the target itself depends on everyone's choices.

2.2 Levels of Reasoning

To find a good strategy, players must think about what others will do. This leads to **iterated reasoning**:

Level	Belief About Others	Choice
Level 0	No strategic adjustment (random/naive)	≈ 50
Level 1	Others are Level 0 (avg ≈ 50)	$\frac{2}{3} \times 50 \approx 33$
Level 2	Others are Level 1 (avg ≈ 33)	$\frac{2}{3} \times 33 \approx 22$
Level 3	Others are Level 2 (avg ≈ 22)	$\frac{2}{3} \times 22 \approx 15$
$\vdots$	$\vdots$	$\vdots$
∞	Infinite mutual reasoning	0

The chain of reasoning:

$$50 \longrightarrow 33 \longrightarrow 22 \longrightarrow 15 \longrightarrow 10 \longrightarrow 7 \longrightarrow \cdots \longrightarrow 0$$



2.3 A Dominance-Style Argument

This section makes the convergence to 0 *rigorous* using a **dominance argument**, not just intuition.

Proposition: No rational player chooses above $\frac{200}{3} \approx 66.67$

Argument: Since all $x_i \in [0, 100]$, we have $\bar{x} \leq 100$. Therefore:

$$T(\mathbf{x}) = \frac{2}{3}\bar{x} \leq \frac{2}{3} \times 100 = 66.\overline{6}$$

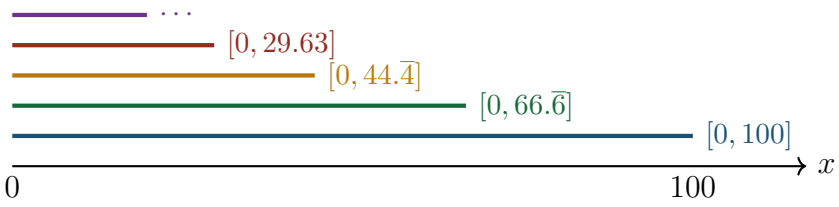
So no matter what others do, the target is *never* above $66.\overline{6}$. Any choice $x_i > 66.\overline{6}$ is always farther from the target than $x_i = 66.\overline{6}$. Hence choosing above $66.\overline{6}$ is **dominated** – a rational player eliminates it.

2.3.1 Iterating the Argument (IESDS)

Once all rational players restrict to $[0, 66.\overline{6}]$, we can apply the argument again:

$$\begin{aligned}
[0, 100] &\xrightarrow{\text{Round 1}} \left[0,\ \tfrac{2}{3} \times 100\right] = [0,\ 66.\overline{6}] \\
&\xrightarrow{\text{Round 2}} \left[0,\ \tfrac{2}{3} \times 66.\overline{6}\right] = [0,\ 44.\overline{4}] \\
&\xrightarrow{\text{Round 3}} \left[0,\ \tfrac{2}{3} \times 44.\overline{4}\right] = [0,\ 29.63] \\
&\xrightarrow{\text{Round } k} \left[0,\ 100 \cdot \left(\tfrac{2}{3}\right)^k\right] \\
&\xrightarrow{k \to \infty} \{0\}
\end{aligned}$$

After infinitely many rounds, *only 0 survives*. This process is called **Iterated Elimination of Strictly Dominated Strategies (IESDS)**.



2.4 What Assumptions Are Used?

Key Insight: The Two Critical Assumptions

The conclusion "all players choose 0" relies on *two* distinct assumptions:

A1. Rationality: Each player avoids dominated strategies.

A2. Common Knowledge of Rationality (CKR): *I am rational. I know you are rational. I know that you know that I am rational. I know that you know that I know ...* and so on, to all orders.

Rationality alone is insufficient – you need CKR to justify each round of elimination.

Note

In real-world experiments with first-time players, the winning number is typically **much larger than 0** (often around 20–35). This is not a failure of the theory – it reveals that real people apply only 1–3 levels of reasoning, and CKR does not hold in practice.

3 The Keynesian Beauty Contest Analogy

John Maynard Keynes drew a famous analogy between financial markets and newspaper beauty contests:

Example: Keynes' Analogy (1936)

In 1930s newspaper contests, readers voted on which face was prettiest from a set of photographs. The prize went to whoever correctly predicted the *average opinion* – not who actually had the prettiest face.

Keynes argued financial speculation works similarly:
- It is not about picking the stock that *you* think is best.
- It is about predicting what *the market* will think is best.
- Or even better: predicting what the market thinks the market will think is best.
- This leads to recursive, self-referential reasoning.

4 Generalization: The p-Beauty Contest

Definition: Generalized p-Beauty Contest

Replace $\frac{2}{3}$ by a parameter $p \in (0, \infty)$. Players choose $x_i \in [0, 100]$ and the target is $T_p(\mathbf{x}) = p\bar{x}$.

Case	Prediction
$0 < p < 1$	**Unique equilibrium**: all choose $x_i = 0$. IESDS converges to 0.
$p = 1$	**Every common choice** is an equilibrium (if all choose the same c, target is c).
$p > 1$	Reasoning moves *upward*. Boundary effects at 100.

5 Sequential Version

Example: Sequential Beauty Contest

Now suppose three players move **in order**:

Step 1. Player 1 chooses x.
Step 2. Player 2 observes x and chooses y.
Step 3. Player 3 observes (x, y) and chooses z.

The target is still $\frac{2}{3} \cdot \frac{x+y+z}{3}$.

Key question: Does observing others' choices change the outcome? Does the equilibrium still predict 0?

(This preview leads into the study of **extensive-form games** and **backward induction**, covered in later lectures.)

6 Summary & Key Takeaways

Key Takeaway

1. Game theory studies **strategic settings** where each player's payoff depends on others' actions – not just optimization.
2. The beauty contest game illustrates that rational behavior requires not just rationality, but **common knowledge of rationality (CKR)**.
3. The solution concept used here is **IESDS** (Iterated Elimination of Strictly Dominated Strategies): repeatedly remove strategies that are always worse than some other strategy.

4. IESDS under CKR predicts choosing 0, yet experiments show most people choose between 20–35 – indicating limited levels of strategic reasoning in practice.
5. Keynes' analogy: financial markets involve not just valuation of assets, but *predicting others' predictions* – a recursive, game-theoretic process.

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